

Reviewer Pack — Minimal Order of Alexander Modules and Communication Complex...

Entry 3431a0a96113 · INCONCLUSIVE

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Minimal Order of Alexander Modules and Communication Complexity Rank

Entry ID: 3431a0a96113 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 18:02:28 UTC

1. Conjecture

Field A (mathematical branch): Alexander Module Theory **Field B** (complexity object): Communication Complexity

Statement:

For every k -communication protocol φ with n participants, the minimal order of an Alexander module associated with the interaction graph of φ is $O(\log(n/k))$.

Rationale (proposer's reasoning):

The Alexander module theory provides a way to encode geometric and topological properties in algebraic terms. The conjecture suggests that the structure of communication complexity can be captured by these algebraic invariants, potentially revealing underlying topological aspects of distributed computations. If true, it would offer a new perspective on complexity lower bounds.

Taxonomy category: Alexander_Module_Theory (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f5303c9b26610129

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal order of an Alexander module associated with a k -communication protocol's interaction graph is considered supported if, across all $n \leq 40$ participants, the mean order $|O(M)|$ satisfies $|O(M)| \leq 1.5 * \log(n/k)$ AND no seed produces $|O(M)| > 2 * \log(n/k)$. The conjecture is falsified if any seed produces $|O(M)| > 2 * \log(n/k)$ or if the mean order $|O(M)| > 1.5 * \log(n/k)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Alexander module theory AND communication complexity rank - minimal order Alexander modules AND interaction graph communication complexity - communication protocol complexity AND Alexander module order $\log(n/k)$

Top relevant hits considered: - [<http://arxiv.org/abs/math/9806161v1>] Chern Classes in Alexander-Spanier Cohomology - [<http://arxiv.org/abs/1403.8106v1>] Recent advances on the log-rank conjecture in communication complexity - [<http://arxiv.org/abs/1102.2932v2>] Applications of Monotone Rank to Complexity Theory - [<http://arxiv.org/abs/0706.2499v1>] Alexander polynomials: Essential variables and multiplicities - [<http://arxiv.org/abs/1705.01819v1>] On quantum cohomology of Grassmannians of isotropic lines, unfoldings of A_n -singularities, and Lefschetz exceptional - [<http://arxiv.org/abs/alg-geom/9703030v1>] Alexander Invariants of Complex Hyperplane Arrangements

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_protocol(n, k):
        protocol = {}
        for i in range(k):
            participants = random.sample(range(1, n+1), 2)
            if tuple(participants) not in protocol:
                protocol[tuple(participants)] = []
            protocol[tuple(participants)].append(random.choice(['send', 'receive']))
        return protocol

    def construct_interaction_graph(protocol):
        graph = {}
        for participants, actions in protocol.items():
            for action in actions:
                if participants[0] not in graph:
                    graph[participants[0]] = set()
                if participants[1] not in graph:
                    graph[participants[1]] = set()
                graph[participants[0]].add(participants[1])
```

```

        graph[participants[1]].add(participants[0])
    return graph

def compute_alexander_module_order(graph):
    # Simplified Alexander module order computation for demonstration
    # This is a placeholder and should be replaced with actual Alexander module computation
    n = len(graph)
    if n == 0:
        return 1
    return n * (n - 1) // 2

def log_base(x, base):
    return math.log(x) / math.log(base)

max_n = 40
total_order = 0
instances_tested = 0

for n in range(5, max_n + 1):
    for k in range(2, min(n, 6)): # Ensure at least 2 participants per protocol
        protocol = generate_random_protocol(n, k)
        graph = construct_interaction_graph(protocol)
        order = compute_alexander_module_order(graph)
        if order <= 0:
            continue
        total_order += order
        instances_tested += 1

mean_order = total_order / instances_tested if instances_tested > 0 else 0
conjecture_holds = all(mean_order <= 1.5 * log_base(n / k, 2) for n in range(5, max_n + 1) for k in range(2, min(n, 6)))

return {
    "metric_name": "minimal_order",
    "metric_value": mean_order,
    "instances_tested": instances_tested,
    "n_max": max_n,
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else "mapping_undefined"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_order = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_order} std=0.0 support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results) and support_fraction >= 0.8:

```

```

print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={seeds[next]}")
else:
    print(f"RESULT: INCONCLUSIVE reason=insufficient_evidence n_tested={len(seeds)}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

r', 'metric_value': 15.65034965034965, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 15.727272727272727, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.783216783216783, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.433566433566433, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.27972027972028, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.3986013986014, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.181818181818183, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.3006993006993, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 15.944055944055943, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.454545454545453, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'minimal_order', 'metric_value': 16.846153846153847, 'instances_tested': 143, 'n_max': 40, 'conjecture_holds': False}
RESULT: INCONCLUSIVE reason=insufficient_evidence n_tested=30

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test only considers up to 40 participants ($n_{\text{max}} = 40$) and tests for a very limited range of k values (2 to $\min(n, 6)$). This is insufficient to validate the conjecture over a wide range of parameters. The metric does not scale trivially with n , but the small size of n tested may lead to false negatives.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that the conjecture does not hold for all instances tested within the range of $n \leq 40$ participants and a limited range of k | next: Further investigation is needed to determine if the conjecture holds for a wider range of parameters. Increase the number of participants and test over a broader range of k values to validate the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14849
2	preregistration	ollama_remote	glm4:latest	0	0	9957
3	novelty	ollama_remote	glm4:latest	0	0	8243
4	novelty	ollama_remote	glm4:latest	0	0	19243
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15132
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	23766

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15445
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17947
9	critic	ollama_remote	glm4:latest	0	0	18082
10	judge	ollama_remote	glm4:latest	0	0	14416

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 157080 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/3431a0a96113.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/3431a0a96113.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/3431a0a96113.tar.gz (if generated)